



UNIVERSITY OF MALAWI - CHANCELLOR COLLEGE

Faculty of Science

Department of Mathematical Sciences

BSc. , BEd.(Science) & BSoc - Year Two

End of Semester Two Supplementary Examination

MAT 221: Calculus II

6th December, 2014	Time Allowed: 3 hours
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Instructions

1. This paper contains **FIVE** questions. **PLEASE CHECK**.
2. Attempt **ALL** questions.
3. Justify your working. Unsupported answers will lead to loss of marks.
4. Use of non-programmable calculators is allowed.
5. Write clearly and neatly.
6. Begin each question on a **new page**.

(Do not start answering the questions until you are told to do so)

Question One [25 Marks]

(a) Differentiate the following functions with respect to v ;

(i) $f(v) = 4v \sinh(3v^2 - 1)$. [3 Marks]

(ii) $g(v) = v^v$. [4 Marks]

(b) Use $x = \cosh u$, to show that $\int \frac{dx}{\sqrt{x^2 - 1}} = \cosh^{-1} x + C$ for $x > 1, u > 0$. [5 Marks]

(c) Evaluate the integral $\int e^{-2t} \cosh(2t) dt$. [3 Marks]

(d) Find the length of the arch, $x = a \cos t, y = a \sin t$ for $0 \leq t \leq 2\pi$. [5 Marks]

(e) Find the area under one arch of cycloid, $x = r(\theta - \sin \theta), y = r(1 - \cos \theta)$ for $0 \leq \theta \leq 2\pi$ about the x -axis. [5 Marks]

Question Two [16 Marks]

(a) Convert the rectangular coordinates $(-2, 2\sqrt{2})$ into polar coordinates. [4 Marks]

(b) Find a polar equation of the hyperbola given by $x^2 - y^2 = 25$. [3 Marks]

(c) Sketch and provide a name of;

(i) $r = -1 - \cos \theta$. [4 Marks]

(ii) $r = \frac{12}{2 + 4 \sin \theta}$. [5 Marks]

Question Three [17 Marks]

(a) Evaluate the following limits;

(i) $\lim_{x \rightarrow 0^+} x \ln x$. [3 Marks]

(ii) $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^{5x}$. [5 Marks]

(b) Determine whether the following improper integrals converge or diverge;

(i) $\int_{\infty}^2 \frac{1}{(x-1)^2} dx$. [4 Marks]

(ii) $\int_0^9 \frac{dx}{(x-1)^{2/3}}$. [5 Marks]

Question Four [24 Marks]

(a) Determine whether the following sequences converge or diverge.

(i) $\left\{ \frac{1}{4}n^2 - 1 \right\}$. [4 Marks]

(ii) $\left\{ \frac{5n}{e^{2n}} \right\}$. [5 Marks]

(b) Use the *root test* to determine the convergence or divergence of the

series $\sum_{n=1}^{\infty} \frac{n^3}{2^n}$. [5 Marks]

(c) Use the *test of divergence* to show that the series $\sum_{n=1}^{\infty} \frac{n^3}{5n^3 + 4}$ diverges. [5 Marks]

(d) Use the *integral test* to determine whether the series $\sum_{n=1}^{\infty} \frac{1}{n^2 + 1}$ converges or not. [5 Marks]

Question Five [18 Marks]

(a) Find a power series representation for xe^{-2x} . $\left[\text{Hint: } e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \right]$. [5 Marks]

(b) Use the first three terms in (a) to estimate $\int_0^1 xe^{-2x} dx$ to three-decimal-place accuracy. [3 Marks]

(c) Find the general solution to the equations;

(i) $\frac{dy}{dx} = 2x$ for $x > 0$. [2 Marks]

(ii) $\frac{dy}{dx} = \frac{2y}{x}$ for $x > 0$. [3 Marks]

(d) Show that $\overline{z + w} = \bar{z} + \bar{w}$ for $z, w \in \mathbb{C}$. [5 marks]

END OF QUESTION PAPER